

TONY HSU TAI

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EDUCATION

Northeastern University

Boston, MA

M.S. in Computer Science – GPA: 3.97/4.00

Sep 2024 – May 2027

- Coursework: Data Structures and Algorithms, Foundations of Artificial Intelligence, Scalable Distributed Systems

Northeastern University

Boston, MA

Accelerated BS/MS Program in Industrial Engineering – GPA: 3.75/4.00

Sep 2016 – May 2021

PROFESSIONAL EXPERIENCE

Software Developer Intern

May 2026 – Present

Sentinel Group

Wakefield, MA

- Enhanced the document-generation API with built-in chart generation by **implementing server-side rendering using Chart.js and Puppeteer**, allowing PDF templates to render charts directly from data rather than uploaded images
- Improved operational flexibility and streamlined production debugging by introducing **runtime-configurable log levels through environment variables**, allowing logging verbosity to be adjusted without redeployments.
- Implemented an event-driven payroll **notification pipeline backed by Amazon SQS and a PostgreSQL persistence table**, ensuring durable state management and fault-tolerant notification delivery.

Operations Manager

Sep 2021 – Jun 2024

Boston Microgreens, LLC

Boston, MA

- Built and maintained an internal scheduling system in Excel to **automate seeding/germination/harvest planning for 50+ weekly orders**, improving reliability of daily operations over 2+ years.
- Designed a germination tracking system in Notion using relational databases and computed fields, eliminating manual checks and **reducing task time by 63%**.
- Improved KPI and profitability metrics to increase reporting accuracy, **revealing under-performing crops** and helping the plant science team focus their testing efforts on those varieties.

PROJECTS

🔗 AI Financial Analysis Platform (Personal) | TypeScript, Node.js, PostgreSQL, React, Gemini API Dec 2025 – June 2026

- Built a multi-stage bank statement **ingestion pipeline that extracts account metadata and transaction records** from PDF statements using layered regex, asynchronous normalization and persisting data in PostgreSQL.
- Engineered a three-stage transaction classification system that uses a 400+ merchant database, 40+ regex-based categorization rules, and a Gemini LLM fallback, **reducing API usage while maintaining classification accuracy**.
- Built a natural language financial chat that translates user queries into validated parametrized SQL through **schema-aware LLM prompting**, returning structured JSON results that are transformed into user readable responses.

🔗 Agricultural Operations System (Academic) | Python, TypeScript, MySQL, React, FastAPI Nov 2025 – Dec 2025

- Designed a full-stack operations platform using React and FastAPI aimed at replacing spreadsheet-based workflows with a centralized **MySQL data model**, improving planning reliability, data integrity and crop traceability.
- Implemented **JWT-based authentication** and structured request validation with Pydantic to enforce clear REST API contracts and reduce backend edge-case failures.
- Built a multi-page React frontend with entity-focused views that present relevant data through tables or cards, supporting **full CRUD operations for core domain entities**.

🔗 Pokémon Battle Game (Personal) | TypeScript, React, Socket.IO, Express.js, Node.js Jul 2025 – Sep 2025

- Built a real-time multiplayer system using Socket.IO with a defined **client-server event protocol** for matchmaking, turn submission, and synchronized battle state updates.
- Implemented a battle simulation engine that handles Pokémon type effectiveness, move priority, and damage variations, keeping game logic separate from Socket.IO event handling to **maintain consistent game state across clients**.
- Integrated **Motion animations** for Pokémon spawns, attacks and health changes to create game-like visual feedback.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL, HTML/CSS

Frameworks & Libraries: React, FastAPI, NestJS, Express.js, Tailwind CSS

Databases & Tools: MySQL, PostgreSQL, Docker, Git, GitHub, Linux, Postman, AWS, Gemini LLM, Claude Code

Spoken Languages: English (native), Spanish (native), Mandarin (beginner)